Reg. No. [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [13AE51]

FIFTH SEMESTER B.TECH EXAMINATION, DECEMBER – 2015

PROPULSION – I

Time: 3 Hours **Maximum Marks:70**

Instructions:

1. Answer any **FIVE** full Questions.
2. Missing data may be suitably assumed.

1. The following data apply to a twin spool turbofan engine with the fan driven by the LP turbine and the compressor by the HP turbine. Separate hot and cold nozzles are used.

Overall pressure ratio	=	19.0
Fuel pressure ratio	=	1.65
Bypass ratio	=	3.0
Turbine inlet temperature	=	1300K
Air mass flow	=	115Kg/s

Calculate the pressure at inlet of the primary nozzle, if the ambient pressure and temperature are 1 bar and 277K. Heat of reaction of the fuel is 43MJ/Kg. **14**

2. **a.** A gas turbine power plant operation on an ideal Brayton cycle has a pressure ratio of 8. The gas temperature is 300K at the compressor inlet and 1300K at the turbine inlet. Assuming a compression efficiency of 80% and a turbine efficiency of 85%, determine:

i. The thermal efficiency

ii. The turbine exit temperature use the air standard assumptions. **12**

b. A turbofan engine has a bypass ratio of 5 and a total mass flow rate of 120kg/s. The mass flow rate through the bypass duct is :(Choose the correct answer from below) **02**

i. 20kg/s **ii.** 100kg/s **iii.** 120kg/s **iv.** 600kg/s

3. **a.** Describe the function of an aircraft air intake. **04**

b. Explain the different types of supersonic air intakes with neat diagrams. **10**

4. **a.** Briefly explain the combustion process in gas turbine combustors. **06**

b. Explain the different zones of a combustion with the help of a simple sketch. **08**

5. **a.** Plot the variation of mass flow rate with pressure ratio in a convergent nozzle. **04**

b. Air enters a nozzle operating at steady state at 3 bar, 440K, with a velocity of

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145m/s, and expands isentropically to an exit velocity of 460m/s, determine:

i. The exit pressure in bar

ii. The ratio of the exit area to the inlet area

iii. Whether the nozzle is convergent, divergent or convergent-divergent. **06**

c. What is the different between under expansion and over expansion conditions of a Nozzle? **04**

6. a. Explain the working principle of a typical centrifugal compressor with a neatly labelled diagram. **08**

b. What are the differences between axial flow and centrifugal flow compressors? **06**

7. a. Write short notes on:

i. Stage loading ii. Degree of reaction **4+4**

b. Draw the velocity triangle at the rotor inlet of an axial turbine stage. **04**

c. What is the difference between the two types of turbine stages? **02**

8. Explain all the parameters which determine the performance of an aircraft engine. **14**